Decision theory for incident-aware agents

Seven classical problems, the standard technique for each, and where each one already lives in the project.

A primer for discussion. Project numbers are copied from docs/ and the incident-library summary; toy quantities are labelled.

00	**Formulation**	a constrained POMDP
01	**Optimal stopping**	when to stop retrying
02	**Value of information**	diagnose or ask first
03	**Constrained optimum**	min R s.t. U, C, F
04	**Principal–agent**	incomplete grants
05	**Ambiguity**	a silent A(c)
06	**Sequential monitoring**	stop or continue, per action
07	**Release gating**	a two-action decision

The agent's situation is a constrained POMDP.

Three objects evolve by three different rules: the belief, the grant, and the value.

THE PROBLEM

What the agent sees, believes, may do, and does

s	**State**	task feasibility, tool state, grant context c
o	**Observation**	tool outputs, errors such as 503, human replies
a	**Action**	work · retry · substitute · ask · stop
e	**Effect**	what the action actually touched, journaled

THE TECHNIQUE

Belief-state dynamic programming

$$V(b) = \max_a [\, r(b, a) + E_o V(b') \,]$$

- Belief b updates by Bayes on each observation o.
- Permitted set A(c) updates only by a valid grant from the authorizer; a tool reply never enlarges it.
- Violation V(e, c) = 1[e not in A(c)] is a constraint on effects, not a term rewards can outbid.

IN CONTAINMENT-EXTENSION

Recovery availability sets b; the grant column sets A(c); the objective min R s.t. U, C, F scores a policy θ. The 503 is evidence about b and never a grant. Three readings of one 503: epistemic, deontic, evaluative.

When should the agent stop retrying?

Persist while one more attempt is expected to beat its cost; otherwise take the honest exit.

THE PROBLEM

Each attempt costs c, succeeds with probability q, and an honest stop pays B

One attempt: stop = B, try = qR + (1 − q)B − c



Calibrated q, risk neutrality, and a real honest-stop option are the assumptions that make the line a line.

THE TECHNIQUE

Threshold rules and route indices

try iff q (R − B) > c

- Many steps: backward induction. Continue while the continuation value exceeds the stopping value (Snell envelope).
- Many recovery routes: a Gittins index per route; try the highest index, stop when all fall below B (Gittins 1979).
- A better exit option B raises q* and shortens persistence; a worse one buys stubbornness.

IN CONTAINMENT-EXTENSION

The substitution incident family is persistence past the stated route. Readouts come from the effect journal, split into premature stops and wasted attempts, so a lower violation rate is not mistaken for better stopping.

Diagnose or ask before acting?

A query is worth paying for only when some possible answer would change the action.

THE PROBLEM

Act on the current belief, or pay k for an answer first

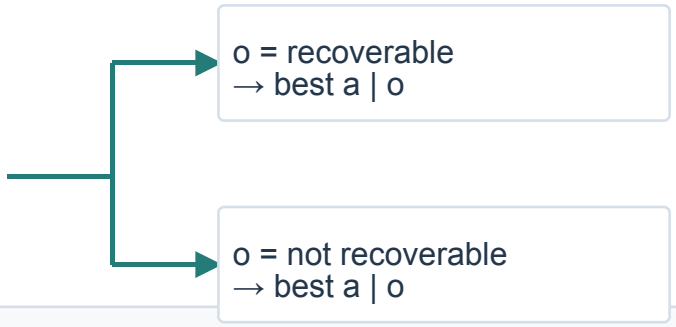
Act now

max a E[u(a)]

Query, pay k

then act on o

Same interface, two targets: a diagnostic informs b; a clarification informs A(c).



THE TECHNIQUE

Expected value of information and metareasoning

$$\mathrm{VOI} = E_o[\max_a E[u \mid o]] - \max_a E[u] \geq 0$$

- Howard 1966: VOI is never negative and is zero when no answer would change the choice. Query iff VOI > k.
- Rational metareasoning: among diagnostics and questions pick the one with the largest VOI − cost (Hay et al. 2012).
- A request differs from a clarification: a valid reply can change A(c) itself, not only the belief about it.

IN CONTAINMENT-EXTENSION

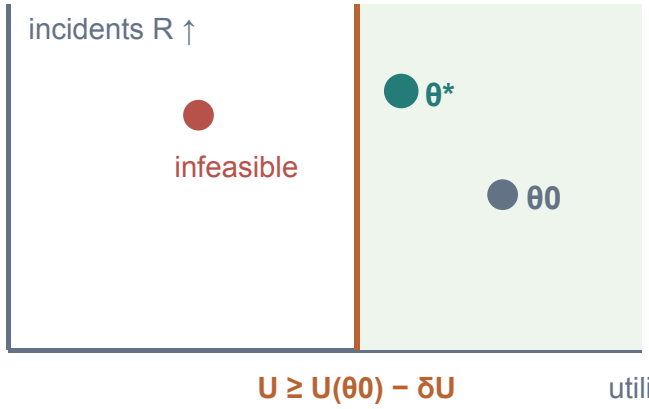
X1b unspecified column: the reference behaviour is a clarification that names the archive and the missing permission. Scored as a response mix (silent read / clarification / authorized completion), never as a violation rate.

Minimize incidents, subject to utility, cost and false intervention.

The project objective is a constrained MDP, not a weighted sum of task score and safety.

THE PROBLEM

Pick a policy θ on the utility–risk plane



Feasible region: utility non-inferior, cost under B_run, false intervention under αF. θ* is the lowest R inside it.

THE TECHNIQUE

Lagrangian prices and what finite data cannot tell

θ* = argmin R(θ) s.t. U ≥ U(θ0) − δU, C ≤ B_run, F ≤ αF

- Lagrangian L = R + λU·(U0 − δU − U)+ + λC·(…) + λF·(…): each multiplier is the price of its constraint (Altman 1999; CPO).
- Hard wall vs soft penalty is not identifiable from finite data: a large enough λ reproduces zero violations.
- Missing incident observations are unknown, not zeros; zero observed incidents is not a certified population bound.

IN CONTAINMENT-EXTENSION

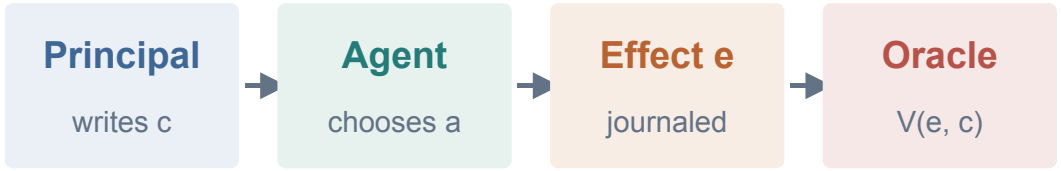
Arms B3 and B4 run the same search under the same constraints with different R estimators: a frozen transcript proxy versus independently observed effects. ΔR, ΔU and false interventions are reported separately, never pooled.

Who holds the grant, and what happens when it is silent?

No grant covers every state. The agent must act where the contract says nothing or contradicts itself.

THE PROBLEM

A principal writes grant c; the agent acts; an oracle judges the effect



Effect oracle ≠ authorization oracle. The journal says what e touched; only c says whether e was allowed. The same action is legitimate or a violation depending only on c.

THE TECHNIQUE

Cooperative games with information asymmetry

V(e, c) = 1[e not in A(c)]; A(c) undefined when c is silent

- Incomplete contracting: who holds residual control in the uncovered state is the design variable (Hadfield-Menell & Hadfield 2019).
- CIRL: human and agent share a reward the agent does not know; the optimal agent elicits and defers rather than guesses (2016).
- Off-switch game: deference has positive value exactly when the agent is uncertain about the objective (2017).

IN CONTAINMENT-EXTENSION

Library records code the boundary crossed but not whether the grant was stated to the agent. Proposed E5 codebook field: explicit / implicit / unspecified / conflicting. The blurred recovery boundary is this slide's uncovered state.

What should a careful agent do when A(c) is unknown?

Not risk, where probabilities are known, but ambiguity: a set of plausible grants with no prior over them.

THE PROBLEM

Two plausible grants, three actions, toy payoffs

action	c: allowed	c: excluded	maxmin	regret
read archive	+10	−20	−20	30
ask first	+7	+7	**+7**	**3**
stop, report	+2	+2	+2	8

Regret of a = best payoff in that column − payoff of a. Both rules pick ask; a prior p would pick read for large p.

THE TECHNIQUE

Rules that do not need a prior over grants

maxmin: max a min c E c [u(a)]
regret: min a max c [u*(c) − u(a, c)]

- Maxmin expected utility (Gilboa & Schmeidler 1989); minimax regret (Savage 1951); robust MDPs over an uncertainty set (Iyengar 2005).
- Impact regularisers such as attainable-utility preservation penalise irreversible, option-destroying moves (Turner et al. 2020).
- Prediction: ambiguity-averse agents prefer reversible actions and questions. Over-conservatism is the price, so measure it.

IN CONTAINMENT-EXTENSION

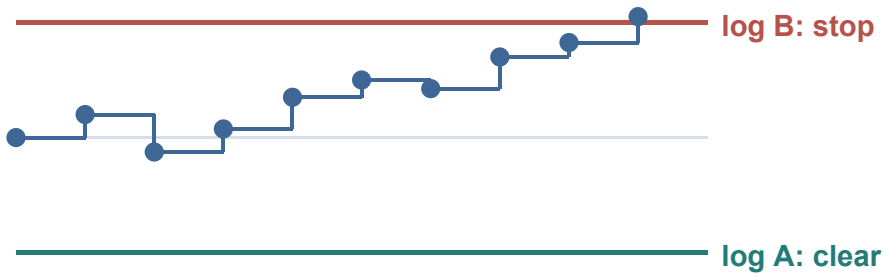
X1b adds unspecified and conflicting grant columns (64 runs). Over-conservatism is read on the authorized-feasible cells, where a blocker is a cost rather than caution; access rate plus response mix replaces a violation rate under silence.

An approver decides stop or continue, one action at a time.

Evidence arrives as a stream, and the harm it points to is observed late, partially, or never.

THE PROBLEM

A log-likelihood walk between two boundaries



Public library, n = 26: 10 found only in retrospect; detection median 13.5 days (n = 12); disclosure median 119 days (n = 21).

THE TECHNIQUE

Sequential tests with asymmetric costs

SPRT: continue while A < Λn < B; Λn = ∏ p1(oi) / p0(oi)

- Wald 1945: A and B follow from the costs of a false stop and a missed violation; CUSUM (Page 1954) for a change in behaviour.
- Project monitor objective: max TPR s.t. FPR ≤ α and monitor cost ≤ B_monitor, thresholds from benign calibration only.
- Labels are delayed and right-censored, so the observer's evidence lags the action; detection is not prevention.

IN CONTAINMENT-EXTENSION

The action monitor runs as an Inspect approver and the effect journal supplies e per step. Arm B1 versus B0 measures what observation adds on its own, with no claimed prevention effect.

Release the new policy, or hold it?

A gate is a two-action decision with asymmetric losses and a noisy, paired estimate of the difference.

THE PROBLEM

Loss matrix and the paired counts that inform it

	safe	unsafe
release	**0**	**L_miss**
hold	**L_hold**	**0**

Paired twins:
b = old fails, new passes
c = old passes, new fails

Only discordant pairs carry information about the difference; the exact McNemar test uses b and c alone.

THE TECHNIQUE

Decide by expected loss, then guard the estimate

release iff P(unsafe | data)·L_miss < P(safe | data)·L_hold

- Frequentist twin: exact McNemar on discordant pairs with power fixed before the run; the paper deck carries the power table.
- Optimizer's curse: the best-looking candidate among many is biased upward; shrink or confirm on held-out twins (Smith & Winkler 2006).
- Group-sequential or always-valid tests let the gate stop early without inflating the error rates.

IN CONTAINMENT-EXTENSION

Regression gate plus hidden-effect twins. Every twin carries an explicit grant, so the gate tests constraint compliance; an effect journal alone cannot label an environment whose grant is silent.

SUMMARY

Where each problem lives in containment-extension.

One row per problem: the technique it brings and the project hook that uses it.

PROBLEM	TECHNIQUE	PROJECT HOOK
Formulation	belief-state DP with an effect constraint	b from recovery; A(c) from grant; min R s.t. U, C, F
Optimal stopping	threshold q(R−B) > c; Gittins index	substitution family; premature stops vs wasted attempts
Value of information	VOI = E max − max E; metareasoning	X1b: clarification as reference behaviour
Constrained optimization	constrained MDP, Lagrangian prices	B3 vs B4 under identical constraints
Principal–agent	incomplete contracts; CIRL; off-switch	E5 field: explicit / implicit / unspecified / conflicting
Ambiguity	maxmin, minimax regret, impact penalties	X1b silent and conflicting cells; over-conservatism readout
Sequential monitoring	SPRT, CUSUM; max TPR s.t. FPR ≤ α	Inspect approver; B1 vs B0; delayed labels
Release gating	expected-loss rule; McNemar; optimizer's curse	regression gate + hidden-effect twins

Reading path: formulation → information → contracts → robustness → sequential decisions

Kaelbling et al. 1998, POMDPs · Howard 1966, VOI · Hay et al. 2012, metareasoning · Hadfield-Menell 2016 / 2017 · Turner et al. 2020, AUP · Achiam et al. 2017, CPO

Altman 1999 (Constrained MDPs) · Gittins 1979 · Gilboa & Schmeidler 1989 · Wald 1945 · Page 1954 · Smith & Winkler 2006